Addition of Two Binary Numbers

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1 Introduction

Arithmetic operations on the computer are done using the binary numbering system in which each digit is either 0 or 1. Binary addition is the basic operation used in arithmetic circuits. Digital circuits known as Adders are specifically made for the purpose of doing binary additions.

2 Binary Number System

The binary number system is a base-2 number system that uses only two digits: 0 and 1. Binary representation is ideal for digital hardware because electronic devices such as transistors can easily distinguish between two voltage levels, representing logic LOW (0) and HIGH (1).

Every operation inside a CPU, GPU, memory controller, and cryptographic processor relies upon binary arithmetic.

3 Binary Addition Rules

Binary addition follows four simple rules:

- 0 + 0 = 0
- 0 + 1 = 1
- 1 + 0 = 1
- 1 + 1 = 10 (Sum = 0, Carry = 1)

These rules form the basis of all binary arithmetic operations.

4 Half Adder

4.1 Definition

A **Half Adder** is the simplest combinational logic circuit used for adding two single-bit binary numbers.

It accepts the following inputs:

- Input A
- Input B

It produces the following outputs:

- Sum (S)
- Carry (C)

A Half Adder does not accept an incoming carry from a previous stage.

4.2 Logic Diagram Explanation

A Half Adder consists of two logic gates:

- XOR Gate $\rightarrow$ Sum
- AND Gate $\rightarrow$ Carry

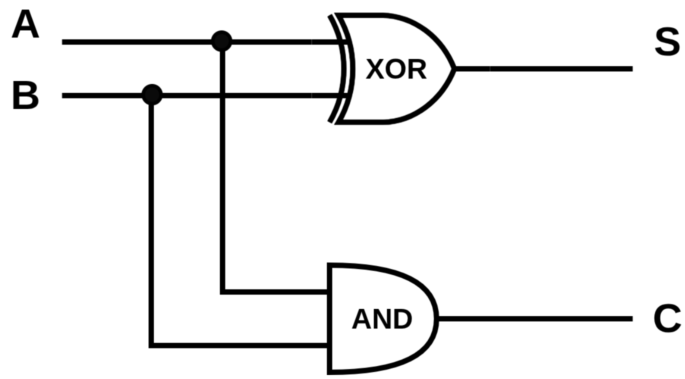

Figure 1: Logic Diagram of Half Adder

The XOR gate generates the sum because XOR outputs 1 only when exactly one input is 1.
The AND gate generates the carry because a carry occurs only when both inputs are 1.

4.3 Truth Table

A	**B**	**Sum**	**Carry**
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Table 1: Truth Table of Half Adder

4.4 Boolean Expressions

The Boolean expressions for a Half Adder are:
Sum (S):

$$S = A \oplus B$$

Expanded form:

$$S = \overline{A}B + A\overline{B}$$

Carry (C):

$$C = A \cdot B$$

5 Full Adder

5.1 Definition

A Full Adder is a combinational logic circuit that adds three binary inputs.

Inputs:

- Input A
- Input B
- Carry Input C_{in}

Output:

- Sum
- Carry Output C_{out}

Unlike the Half Adder, the Full Adder can process carries generated from previous stages, making it suitable for multi-bit arithmetic.

5.2 Logic Diagram Explanation

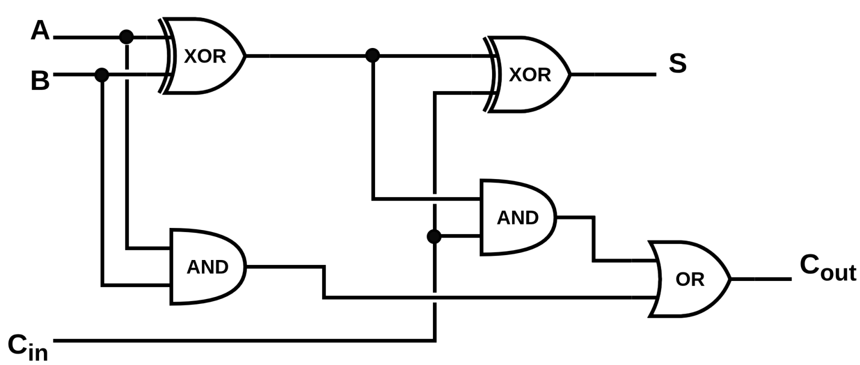

Figure 2: Logic Diagram of Full Adder

The Full Adder can be implemented using two Half Adders and one OR gate. The first Half Adder adds the inputs A and B. The second Half Adder adds the intermediate sum with the carry input (C_{in}). Finally, the OR gate combines the carry outputs to produce the final carry (C_{out}).

5.3 Truth Table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Table 2: Truth Table of Full Adder

5.4 Boolean Expressions

The Boolean expressions for a Full Adder are:

Sum (S):

$$S = A \oplus B \oplus C_{in}$$

Carry (C_{out}):

$$C_{out} = AB + AC_{in} + BC_{in}$$

An equivalent expression for the carry output is:

$$C_{out} = AB + (A \oplus B)C_{in}$$

6 Ripple Carry Adder

A Ripple Carry Adder (RCA) is a combinational circuit used to add two multi-bit binary numbers. It is formed by connecting multiple adder circuits in sequence. The carry generated by one stage is passed to the next stage until the final result is obtained.

There are two common implementations of a Ripple Carry Adder:

1. **One Half Adder and ((n-1)) Full Adders:**
   Used when there is no initial carry input. The Half Adder adds the least significant bits, and the remaining bits are added using Full Adders.

2. **(n) Full Adders:**
   Used in most practical digital systems. The first Full Adder receives an initial carry input ($C_{in} = 0$), making the design simple and uniform.

7 Example: Addition of Two 4-bit Binary Numbers

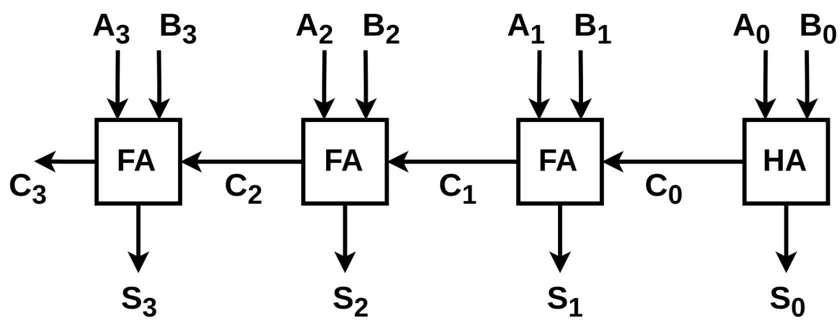

Figure 3: 4-Bit Ripple Carry Adder

7.1 Working:

1. The Half Adder adds the first pair of bits ((A_0) and (B_0)) and generates the first sum and carry.
2. The carry is passed to the next Full Adder.
3. Each Full Adder adds two input bits and the carry from the previous stage.
4. This process continues until all four bits are added.
5. The final carry becomes the most significant bit (MSB) of the result.

Since the carry moves from one stage to the next, it is called a Ripple Carry Adder. This design is simple but becomes slower for larger binary numbers because each stage must wait for the previous carry.

8 Conclusion

Binary addition is one of the fundamental operations in digital systems. The Half Adder performs the addition of two single-bit numbers, while the Full Adder can also process a carry input. By connecting multiple Full Adders, a Ripple Carry Adder can add multi-bit binary numbers. These adder circuits are widely used in processors, arithmetic logic units (ALUs), and cryptographic hardware to perform efficient arithmetic operations.

9 References

1. Class Notes – Computing System I.
2. M. Morris Mano, *Digital Design*, 5th Edition.
3. Images adapted from standard Digital Logic resources.